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MILITARY SPECIFICATION

REPEATER, TELEPHONE (TELEPHONE REPEATER TP-14( ))

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers an adjustable-gain, two-way voice frequency telephone repeater used as an amplifier at intermediate or terminal points on two-wire telephone lines. The repeater may be operated from any of three power sources which are 115 or 230 volts, 50 to 60 c.p.s., single phase alternating current; a 12 volt storage battery; or 126 volt and 12 volt dry batteries.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and drawings of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

QQ-S-571 — Solder; Soft (Tin, Tin-Lead, and Lead-Silver).

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636 — Boxes, Fiber.

PPP-T-60 — Tape; Pressure Sensitive Adhesive, Waterproof—for Packaging and Sealing.

MILITARY

MIL-P-116 — Preservation, Methods of.

MIL-B-117 — Bags, Interior, Packaging.

MIL-W-12683 — Welding; Joint Designs, Data for (Structures other than Armor).

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-109 — Inspection Terms and Definitions.

FSC 5965

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MIL-STD-129 — Marking for Shipment and Storage.

MIL-STD-130 — Identification Marking of U.S. Military Property.

MIL-STD-169 — Extreme-Temperature Cycle.

MIL-STD-170 — Moisture Resistance Test Cycle for Ground Signal Equipment.

DRAWINGS

SIGNAL CORPS

SC-DL-39291 — Telephone Repeater TP-14( ).

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from or as directed by the contracting officer. Both the title and identifying number or symbol should be stipulated when requesting copies.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise specified the issue in effect on date of invitation for bids applies.

ASSOCIATION OF AMERICAN RAILROADS

Uniform Freight Classification Ratings, Rules, and Regulations.

(Application for copies should be addressed to the Association of American Railroads, 202 Chicago Union Station, Chicago 6, Illinois.)

AMERICAN TRUCKING ASSOCIATION

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, 1424 16th Street, N. W., Washington 6, D. C.)

3. REQUIREMENTS

3.1 Description. Telephone Repeater TP-14( ) is a portable telephone repeater consisting of amplifiers, balancing networks, and a rectifier unit inclosed in a wooden case. Basically, it is a two wire repeater, capable of four wire operation by simple external wiring changes.

3.2 Construction. Telephone Repeater TP-14( ) shall be constructed in accordance with Drawing SC-DL-39291, and this specification.

3.3 Parts, materials and processes; general.

3.3.1 *Soldering.*

3.3.1.1 *Solder, soft.* Soft solder shall be in accordance with Specification QQ-S-571, and where used for electrical connections shall have a minimum tin content of 40 percent by weight unless otherwise specified.

3.3.1.2 *Flux and cleaning agents.* No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use. Flux for soldering of electrical connection shall be rosin, rosin and alcohol, or rosin and turpentine.

3.3.1.3 *Process.* There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting action, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not include the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area, or damage to the equipment. Insula-

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tion material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.3.2 *Welding.* Welds shall be of ample size. Where tack or spot welds are employed, the number of welds shall be sufficient to provide adequate strength and at least two welds shall be used to hold each welded part. Spot or tack welds in aluminum or magnesium shall be limited to lightly stressed, noncritical structures. Wherever practicable, the welded joint shall conform to Specification MIL-W-12693, and shall be such that grinding on the finished weld will be unnecessary. Surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from rust, scale, paint, grease, and other foreign material.

3.3.2.1 *Process.* Preheating shall be employed where distortion is likely to result from welding. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, and other harmful defects. Where undesirable internal stresses are likely to result from welding, the welded assembly shall be stress-relieved. Inert-gas-shielded arc welding shall be used when practicable for welding of aluminum or magnesium.

3.3.3 *Cleaning.* Soldered parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document, removing all solder particles, flux and other foreign material from components. Welded parts and assemblies shall be cleaned to remove scale, oxidation products and excess flux by sandblasting, wire brushing or other suitable means. Cleaning processes shall have no deleterious effect on the equipment. Corrosive material shall be removed completely before parts are assembled into the equipment.

3.4 Electrical requirements.

3.4.1 *Power supply.* The equipment shall meet the requirements of table I with power sources as specified.

3.4.2 *Return loss for networks.* The return loss of Network 1 and Network 2, over the frequency range of 500 to 2500 c.p.s., shall be not less than 30 db when tested in accordance with 4.7.

3.4.3 *Return loss for building-out capacitors.* The return loss of the building-out capacitors, over the range of 500 to 2500 c.p.s., shall be not less than 28 db, when tested in accordance with 4.8.

3.4.4 *Gain-frequency characteristics.* The insertion gain of the repeater, as measured in accordance with figure 8, shall meet the requirements of the curve in figure 4 and shall be capable of meeting the tests of 4.9.

3.4.5 *Gain switches.* Each step of the gain control switch of each amplifier shall decrease the gain by 2.0 db ± .3 db when rotating the dial from maximum to minimum.

3.4.6 *Gain-load characteristics.* The gain of the repeater shall not vary by more than 0.5 db when tested in accordance with 4.11.

3.4.7 *Singing point.* The repeater shall not "sing" when tested in accordance with 4.12.

3.4.8 *Harmonic distortion.* The harmonic distortion measured at 400 c.p.s. shall not exceed 5 percent at all output levels from 0 dbm up to and including ± 10 dbm.

3.4.9 *Noise level.* The noise level shall not exceed 20 db RN (reference noise level) when tested in accordance with 4.14.

3.4.10 *Impedance.* The impedance of the repeater shall be 600 ohms ± 10 percent over the frequency range of 500 c.p.s. to 2500 c.p.s.

3.4.11 *Twenty-cycle bypass circuit.* The

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efficiency of the twenty-cycle bypass circuit will be determined from figure 9 and shall be tested in accordance with 4.16.

3.4.12 *Circuit loss.* The loss from the monitor/talk binding posts to the line terminals, with the key in the "Talk" position, shall not exceed 11.5 db when measured between 600 ohm resistive terminations as tested in 4.17.

3.4.13 *Longitudinal balance.* The longitudinal balance of the repeater shall be at least 60 db at 1000 c.p.s. when tested in accordance with 4.18.

TABLE I

Operations		Results				
TP-14 voltage selector	Test circuit switch	Input voltage *	Current drain (Input ammeter)	Tube heater voltage	Tube plate voltage (meter G)	Hum level (meter D)
12 V	S_1 Closed S_2 Open S_3 To meter A	11 V d.c. 12 V d.c. 13 V d.c.	1.0 amp max	11 12 13 (Meter A).	115–150	Less than 30 mv.
External	S_1 Open S_2 Closed S_3 To meter A	183 V d.c. 185 V d.c. 187 V d.c. 15 V d.c.	19–20.5 145–155 ma	(Meter A) 12 V d.c.	183 V 185 V 187 V	Less than 30 mv.
115 V	S_3 To meter B	103.5 V a.c. 115. V a.c. 126.5 V a.c.	100 ma max	11–14 (Meter B).	115–150	Less than 30 mv.
230 V	S_3 To meter B	207 V a.c. 230 V a.c. 253 V a.c.	70 ma max	11–14 (Meter B).	115–150	Less than 30 mv.

* Adjust input variable resistor (d.c. voltage) or varine (a.c. voltage) to each value listed.

3.5 **Breakdown voltage.** When inspected in accordance with 4.5 the repeater shall show no visible deterioration and shall meet all specified performance requirements of 3.4.

3.6 **Service conditions.** The repeater shall meet the following service conditions:

3.6.1 *Equipment operating.* The repeater shall meet the requirements of this specification under any probable combination of the following:

(a) *Temperature.* Any ambient temperature in the range of —65°F. to +150°F. (The +150°F. temperature includes effect of sun-load.)

(b) *Relative humidity.* Any relative humidity up to 98 percent.

3.6.2 *Equipment nonoperating.* The repeater shall meet the requirements of 3.6.1 after subjection to any one or more of the following nonoperating conditions:

(a) *Temperature.* Continuous exposure for 4 hours at +160°F. and 4 hours at —80°F., at any orientation.

(b) *Relative humidity.* Relative humidity up to 100 percent, including condensation caused by temperature changes. Following subjection to such conditions and a power-on period of 16 hours at normal room temperature and relative humidity, the equipment shall meet the requirements of 3.6.1.

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(c) *Rain.* Two inches of rainfall per hour and wind at 20 miles per hour. (see 4.21).

(d) *Vibration, bounce and shock.* See 3.7.

3.7 Vibration, bounce and shock.

3.7.1 *Vibration.* Except for internal resonance of specified parts and assemblies, the repeater shall have no mechanical resonance below 55 cycles per second when subjected to test of 4.22.

3.7.2 *Bounce test.* The repeater shall meet full specification performance and show no physical damage except surface abration when subjected to the test of 4.23.

3.7.3 *Shocks bench handling.* The repeater shall meet full specification performance and show no physical damage when subjected to the test of 4.24.

3.7.4 *Shock: drop.* The repeater shall be operable when subjected to the test of 4.25. Minor mechanical damage is permissible.

3.8 Facsimiles. In addition to any facsimiles specified by applicable documents, 4 copies each of the schematic diagram and the wiring diagram shall be submitted for approval to the contracting officer.

3.9 Bounce preconditioning. The repeaters shall be capable of meeting the inspection of section 4 without subsequent processing after subjection to bounce preconditioning. (see 4.4)

3.10 Preproduction samples.

3.10.1 *Quantities.* Two complete repeaters shall be furnished for Government inspection and approval.

3.10.2 *Compliance with specified requirements.*

3.10.2.1 Preproduction samples shall meet specified requirements, shall be made and assembled by tools and methods that will be used for quantity production and shall be accompanied by a statement to that effect.

3.10.2.2 Approval of preproduction samples shall not be construed as a waiver of any specified requirements. After being released to the contractor preproduction samples to be offered as units on contract shall be refabricated by the contractor if necessary to meet specified requirements.

3.10.2.3 When deviation from 3.10.2 is unavoidable, the preproduction samples may be submitted for approval provided that the accompanying statement describes in detail each nonconforming feature, reason therefor, and manner in which it will be corrected in production of equipment on contract.

3.10.3 *Test data.* The preproduction samples shall be accompanied by test data showing compliance with specified performance. The test data shall comprise an engineering report giving test procedure, observations and other data, calculations, test results and essential details of the testing equipment (manufacturer's model serial number, date of calibration, and the like).

3.10.4 *Reference standards.* After preproduction samples have been approved and returned to the contractor's plant, they shall be kept intact in custody of the Government inspector until released by him. They shall be used as reference standards to resolve any differences of opinion regarding interpretation of requirements.

3.11 Technical literature, tools, and running spare parts. Technical literature, tools, and running spare parts shall be furnished as specified by the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on order.

3.12 Identification of product. Identifica-

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tion information to be marked on parts and assemblies shall be as specified in Standard MIL-STD-130 and shall include the part number, nomenclature, prime manufacturer's identification, and special characteristics. Where space does not permit listing all the foregoing, priority shall be in the order listed.

3.12.1 *Serial numbers.* Serial numbers as defined in Standard MIL-STD-130, are further understood as follows:

The serial numbers shall start with Number 1 for each government order number and continue consecutively unless the contracting officer allows or requires that serial numbers shall be continuous for two or more consecutive or concurrent order numbers.

3.13 Workmanship. The equipment shall be manufactured and assembled in a thoroughly workmanlike manner in accordance with the applicable portions of the following paragraphs herein, when inspected in accordance with 4.26.

Construction	—3.2
Soldering	—3.3.1, 3.3.1.1, 3.3.1.2, 3.3.1.3
Welding	—3.3.2, 3.3.2.1
Cleaning	—3.3.3
Facsimiles	—3.8

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection, general.

4.1.1 *Definitions.* Standard MIL-STD-109 applies for definitions of terms used herein.

4.1.2 *Classification of inspection.* Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (see 4.2).

(b) Acceptance inspection of items before they are prepared for delivery. (see 4.3).

(c) Acceptance inspection of preparation for delivery. (see 4.27 and section 5).

4.2 Preproduction inspection.

4.2.1 *Preproduction inspection of equipment.* Preproduction inspection of equipment shall consist of the group A inspection, specified in table III and the preproduction inspection specified in table II. Other nondestructive inspection on preproduction samples may be performed to determine compliance with specified requirements. Preproduction inspection will normally be performed in this order: (1) vibration, (2) bounce, (3) shock, benchhandling and (4) shock, ballistic or drop; other preproduction inspection may precede, follow or be interspersed between the foregoing.

TABLE II. *Preproduction inspection*

Inspection	Requirement paragraph	Test paragraph
Breakdown voltage	3.5	4.5
Temperature	3.6	4.19
Moisture Resistance	3.6	4.20
Rain	3.6.2	4.21
Vibration	3.7.1	4.22
Bounce	3.7.2	4.23
Bench handling	3.7.3	4.24
Drop	3.7.4	4.25

4.3 Acceptance inspection.

4.3.1 *Group A inspection of equipment.* Group A inspection (including sampling) shall conform to table III and Standard MIL-STD-105.

4.3.2 *Disposition of nonconforming product.* Disposition of rejected lots and defective sample units shall be in accordance with Standard MIL-STD-105.

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TABLE III. *Government Group A inspection*

Inspection	Requirements paragraph	Test paragraph	AQL
Visual and mechanical		4.36	Major 4.0 dphu. Minor 15 dphu.
Power supply	3.4.1	4.6	
Return loss for network	3.4.2	4.7	
Return loss for building-out capacitors	3.4.3	4.8	
Amplifier gain	3.4.4	4.9	
Gain switch	3.4.5	4.10	6.5 dphu * for the group.
Gain-load characteristics	3.4.6	4.11	
Singing point	3.4.7	4.12	
Harmonic distortion	3.4.8	4.13	
Noise level	3.4.9	4.14	
Impedance	3.4.10	4.15	
Twenty-cycle bypass circuit	3.4.11	4.16	
Circuit loss	3.4.12	4.17	
Longitudinal balance	3.4.13	4.18	

* All electrical defects are considered major.

4.4 **Bounce preconditioning.** After final assembly, each telephone repeater which will be examined, measured, or tested shall be preconditioned as follows: The repeater shall be placed in its normal operating position on the table of the Package Tester as made by the L. A. B. Corporation, Summit, New Jersey, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a $\frac{1}{32}$-inch-thick strip of material under one corner or edge of the equipment to a distance of 3 inches as the equipment bounces. The repeater shall be subjected to this preconditioning for 1 minute. After bounce preconditioning the equipment shall not be repaired, alined, cleaned or otherwise changed prior to subjection to acceptance inspection.

4.5 **Breakdown voltage.** Electrical insulation of repeaters shall have the minimum breakdown voltage applied as shown in Table IV. After 10 second application of the voltage, the repeaters shall meet the requirements of 3.5.

4.6 **Power supply.** The repeaters shall be tested as shown in figure 1 to show compliance with 3.4.1. The 12 volt application requires a 12 volt storage battery. The external application requires 135 volt and 12 volt dry batteries.

TABLE IV. *Breakdown voltage*

Operating voltage (Note A)	Breakdown voltage (Note B)
30 to 125 volts	500 volts
125 to 500 volts	4 times operating voltage

Note A. Operating voltage shall be maximum instantaneous a.c. plus d.c. during use of the repeater under specified service conditions.

Note B. Minimum breakdown voltage shall be d.c. or peak a.c. If a.c., the waveform shall be approximately sinusoidal and the frequency not over 100 cycles per second.

4.7 **Return loss for networks.** Network 1 and Network 2 shall be tested in accordance with figure 2. The "Standard Network" is a replica of the network used in Telephone Repeater TP-14( ) and is used for test purposes only. The components of the "Standard Network" shall be held to a tolerance of ±0.5%.

4.8 **Return loss for building-out capacitors.** The building-out capacitors shall be tested as in 4.7 to show compliance with 3.4.3.

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4.9 **Amplifier gain.** The gain of each amplifier with gain switch set on maximum gain shall meet the limits listed in table V. The amplifiers shall be tested separately with one turned off while the other is being tested. The change in gain at 1,000 c.p.s. in equalizer position 1 shall not exceed 0.5 db when the output level is raised from 0 to +10 dbm.

TABLE V. *Amplifier gain.*

Equalizer Position	Frequency (c.p.s.)	Limits in db.
1	400	gain at 1,000 c.p.s. −2.0 ± 1.5.
	1000	18.0 ± 2.0
	2000	gain at 1,000 c.p.s. ± 1.0.
	3500	gain at 1,000 c.p.s. −7.0 or more.
2	400	gain at 1,000 c.p.s. −3.0 ± 2.0.
	1000	18.0 ± 2.0.
	2000	gain at 1,000 c.p.s. +2.0 ± 1.5.
	3500	gain at 1,000 c.p.s. −5.0.
3	400	gain at 1,000 c.p.s. −6.0 ± 2.0.
	1000	18.0 ± 2.0.
	2000	gain at 1,000 c.p.s. +3.5 ± 1.5 (Must be higher than in position 2).
	3500	gain at 1,000 c.p.s. −5.0 or more.

4.10 **Gain switches.** The gain control switch of each amplifier shall be tested to show compliance with 3.4.5.

4.11 **Gain-load characteristics.** The gain of the repeater when tested as in figure 5 shall not vary by more than 0.5 db with an input between 0 and —15 dbm at either 1000 or 2000 c.p.s. The gain shall be measured for both directions of transmission.

4.12 **Singing point.** The repeater shall not "sing" (oscillate) when one line and its associated network are properly terminated and the other line open and its associated network shorted. "Singing" shall not occur when the line termination is changed by less than ±5 ohms. The center of the nonsinging wedge (the wedge being defined as the limits of nonsinging and having a width of 10 ohms) shall fall within the limits of 600 ohms ±30 ohms. This measurement shall be made on the completed equipment with each amplifier gain control set on the 18 db step and with repeater turned on.

4.13 **Harmonic distortion.** The repeater shall be tested as shown in figure 6 to show compliance with 3.4.8.

4.14 **Noise level.** The noise level shall be measured with a Western Electric Co. 2–B noise Measuring Set, or equal, adjusted to the FIA line weighting position. After completion of the test for one direction of transmission, as shown in figure 7, the 600-ohm termination shall be changed from terminals 12 and 13 to terminals 4 and 5, and the meter shall be changed from terminals 4 and 5 to terminals 12 and 13. The noise level for both directions of transmission shall show compliance with 3.4.9.

4.15 **Impedance.** The equipment shall be tested to show compliance with 3.4.10.

4.16 **Tests on twenty-cycle bypass circuit.** The repeater shall be tested in accordance with paragraphs 4.16.1 and 4.16.2 and with figure 9 to determine the efficiency of the twenty-cycle bypass circuit. During performance of these tests, the repeater shall be energized and the two amplifiers shall be disabled.

4.16.1 A 20-cycle generator in series with an impedance of 3,000–j3,000 ohms at 20 c.p.s. shall be connected to one pair of line terminals on the repeater. A similar impedance (3,000–j3,000) shall be connected across the other pair of line terminals. The voltage measured across the latter (output) impedance shall be at least ⅛ of the voltage across the generator terminals, as the generator terminal voltage is varied over the range of 20 to 75 volts.

4.16.2 A 20-cycle generator in series with a 125-volt, 15-watt tungsten-filament lamp shall be connected to one pair of line terminals on the repeater. An impedance of 3,000–

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j3,000 ohms at 20 c.p.s. shall be connected across the other pair of line terminals. The voltage measured across the output impedance (3,000−j3,000) shall be at least ½ of the voltage across the generator terminals, as the generator terminal voltage is varied over the range of 70 to 120 volts.

4.17 Circuit loss. The repeater shall be tested as shown in figure 8. After completion of the test for one direction of transmission, the 600 ohm termination shall be changed from 12 and 13 to terminals 4 and 5, and the meter shall be changed from terminals 4 and 5 to 12 and 13. The loss for both directions of transmission shall conform to 3.4.12.

4.18 Longitudinal balance. The repeater shall be tested as shown in figure 10. After completion of the test for one direction of transmission, the signal generator shall be changed from terminals 4 and 5 to terminals 12 and 13, the measuring circuit shall be changed from terminals 12 and 13 to 4 and 5, and the simplex lead shall be changed from terminal SX 12–13 to SX 4–5. The longitudinal balance for both directions of transmission shall conform to 3.4.13.

4.19 Temperature. The repeater shall be subjected to the temperature cycle shown on Standard MIL–STD–169. The tests of 4.6 through 4.18 shall be made at —65°F. and +150°F.

4.20 Moisture-resistance test for equipment.

4.20.1 *Test conditions.*

(a) Do not move repeaters from the humidity chamber for measurements.

(b) Start measurements not more than 5 minutes after power is applied to the repeater. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

4.20.2 *Test procedure.* The repeaters will be tested as follows:

(a) Dry at 130° ± 5°F. for 24 hours.

(b) Condition at 77° ± 5°F. and 40 to 50 percent relative humidity for 24 hours.

(c) Take measurements as specified in 4.20.3, and readjust or realine as necessary to meet full specification requirements.

(d) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Standard MIL–STD–170. The measurements specified in 4.20.3 shall be made during the periods shown on the standard and shall comply with 4.20.3.

(e) After cycling has been completed, condition the repeaters for 24 hours at 77° ± 5°F. and 40 to 60 percent relative humidity. Then adjust for optimum performance, using only those means provided by the repeaters. No repair or replacement of parts shall be made. After adjustment, the repeaters shall meet full specification requirements for those measurements specified in 4.20.3.

4.20.3 *Performance.* Measurements made during test and specified performance during cycling, shall be as specified in 4.6 through 4.18.

4.20.4 *Failure.* If any repeater fails to meet the performance specified in 4.20.3 during cycling, it shall be realined or readjusted once. If the repeater then fails to meet 4.20.3 or fails subsequently during cycling, it does

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not pass the test. In addition, if the repeater fails to meet full specification requirements after conditioning and adjustment, as specified in 4.20.2(e), it does not pass the test.

4.21 Rain test. The repeater shall be tested, step by step, as follows:

4.21.1 Dry at 150°±5°F. for 48 hours.

4.21.2 Condition for 4 hours at 77°±5°F. and 40 to 60 percent relative humidity.

4.21.3 Perform the tests of 4.6 through 4.18.

4.21.4 Place the repeater, in its normal operating position, in a test chamber maintained at 77°±5°F. Allow simulated rainfall to fall on exposed surfaces of the component. The water shall be at a temperature of 65° ±5°F. and shall fall freely at a rate of 2 inches per hour. Maintain this condition for one hour.

4.21.5 With the rainfall continuing at the 2-inch rate, introduce a simulated wind blowing horizontally at 20±5 miles per hour against one side of the repeater. The water shall fall on exposed surfaces and be blown by the wind against one side of the repeater. Maintain this condition for one hour.

4.21.6 Rotate the repeater 90 degrees horizontally, to expose a second side of the repeater to the wind and rain. Maintain this condition for one hour.

4.21.7 Continue as in 4.21.6, until all four sides of the component have been exposed to the wind and rain, each side for one hour.

4.21.8 Immediately after completion of the foregoing, repeat the measurement specified in 4.21.8 without removal from the chamber but with the rain and wind discontinued. Complete the measurements within one hour. The measurements shall meet specified performance.

4.21.9 *Open the case.* There shall be no visible evidence of leakage.

4.22 Vibration test. The repeater shall be tested for resonant frequencies below 55 cycles per second, to determine compliance with 3.7.1 as follows:

(a) The repeater shall be fastened in its normal mounting position on a vibration table that can be controlled within 10 percent of the specified amplitude. The vibration table shall provide approximately sinusoidal vibration.

(b) The repeater shall be vibrated successively in three mutually perpendicular directions that are parallel respectively to the edges of the equipment, over a frequency range of 10 to 55 cycles per second, in 1-cycle-per-second steps. The total excursion shall be constant at 1/64 inch.

(c) Mechanical resonance, if any, of the complete repeater, of subassemblies, and of component parts shall be determined visually by means of a Strobotac, as made by the General Radio Corporation, Cambridge, Massachusetts, or equal, or by energizing the repeater and detecting mechanical vibration through electrical output indications.

(d) If it is not practical to determine completely any mechanical resonances on the assembled repeater unit because of its compactness or the inaccessibility of its subassemblies or component parts, the unit shall be broken down into logical subassemblies for the above vibration test. Such subassemblies shall be secured to the vibration table in the same manner as they are secured in the equipment unit.

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4.23 **Bounce test.** The repeater in its transit case, shall be placed on the table of the Package Tester as made by the L. A. B. Corporation, Summit, New Jersey, or equal and shall be constrained from horizontal motion of more than 2 inches by suitable wooden fences. The package tester, shafts in phase, shall be operated at a speed of 285 revolutions per minute, ±1 percent, for a total of 3 hours. During each ½ hour period of the test, the transit case (with contained equipment) shall rest on a different face.

4.24 **Shock test, bench handling.** The chassis-and-front-panel assembly shall be removed from its inclosure, as for servicing, and packed in a suitable position for servicing on a solid 2-inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing:

(a) Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as an axis for a total of four drops.

(b) Repeat "(a)" with the assembly resting on other faces, until it has been dropped for a total of four times on each face on which the assembly might practically be placed during servicing.

4.25 **Shock test; drop.** The repeater, in its transit case, shall be dropped a distance of 4 feet on each corner, edge, and face (total of 26 drops). The drops shall be made from a quick-release hook; or pendulum tester as made by the L.A.B. Corporation, Summit, New Jersey, or equal. The floor or barrier receiving the impact shall be of 2-inch fir backed by concrete or by a rigid steel frame.

4.26 **Visual and mechanical inspection.** Parts and repeaters shall be examined for workmanship, mechanical fit, loose nuts and screws, and miscellaneous defects. Controls and fastening devices shall be inspected for mechanical operation. Wiring, soldered connections, ground connections, welds, finishes, etc., shall be inspected for workmanship. Clearances, dimensions, and mechanical adjustments shall be measured. Labels and markings shall be inspected for conformance with approved facsimiles, when facsimiles are required. Examination also may be made for other visual or mechanical defects, similar to those described above, that are contrary to specified requirements for the equipment. Visual and mechanical defects shall be classified by the Government inspector as major or minor in accordance with the definitions of Standard MIL-STD-105.

4.27 **Acceptance inspection of preparation for delivery.**

4.27.1 *Preservation and packaging.* Inspection of preservation and packaging shall be as specified in Specification MIL-P-116. Classification of defects shall conform to table VI.

4.27.2 *Packing and marking of exterior containers.* Packing and marking of exterior containers shall be given visual inspection for the defects listed in table VII and to determine conformance with the approved process sheet furnished by the contractor as required by the contract. This inspection shall conform to the Appendix to Standard MIL-STD-105. Inspection Level L-8 shall be used for normal inspection and L-6 for reduced inspection. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R1. The AQL for major defects shall be four percent and the AQL for minor defects shall be ten percent.

4.27.2.1 *Inspection lot.* A lot for visual inspection of the pack shall be all completed

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packs which are identical and are submitted for inspection at one time.

4.27.2.2 *Procedure in case of failure.* If an inspection lot is rejected the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken. If the contractor and the Government inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the contracting officer for resolution.

4.27.2.3 Disposition of nonconforming product. Disposition of rejected lots and defective sample units shall be in accordance with Standard MIL-STD-105.

4.27.3 *Rough handling test.* When rough handling test is required by the contract, the following functional tests shall be conducted to determine freedom from operational malfunction caused by the rough handling:

Paragraph 4.6 through 4.18.

TABLE VI

Classification of Defects for Preservation, Packaging and Marking

Major Defects

1. Use of improper or defective material.
2. Quantity in unit package not as specified.
3. Incorrect packaging method applied.
4. Cushioning or padding omitted.
5. Cushioning inadequate for the physical and mechanical protection of the item.
6. Unsealed, punctured, or improperly sealed barrier bag, wrap, or envelope.
7. Stock number omitted, incorrect, or illegible.
8. Nomenclature omitted, incorrect, or illegible.
9. Marking of quantity of items in package omitted, incorrect, or illegible.
10. Different stock numbered items in same unit package.

Minor Defects

1. Item not properly blocked or braced within the unit package to prevent movement.
2. Packaging material damaged.
3. Conforming wraps are not snug-fitting and contain voids.
4. Any items of marking information other than 7 through 9 omitted, incorrect, or illegible.

TABLE VII

Classification of Defects for Packing and Marking

Major Defects

1. Use of improper or defective material.
2. Quantity in pack not as specified.
3. Gross weight in excess of specified amount.
4. Box closure not as specified.
5. Type, grade, class and style of the shipping container not as specified.
6. Strapping omitted.
7. Strapping inadequate or incorrectly applied.
8. Items not adequately blocked, braced, or cushioned within the shipping container to prevent movement or damage.
9. Shipping documents or packing list omitted.
10. Stock number omitted, incorrect, or illegible.
11. Nomenclature omitted, incorrect, or illegible.
12. Marking of quantity of items in pack omitted, incorrect, or illegible.
13. Destination marking omitted, incorrect, or illegible.
14. Service designation (color marking) omitted.
15. Specified special marking and labeling such as MAP labels, shipment digit markings, etc., omitted, incorrect or illegible.
16. Overseas code marking omitted, incorrect, or illegible.

Minor Defects

1. Unsealed carton.
2. Defective taping or sealing of carton.
3. Any other box defect which may be considered minor by definition of Standard MIL-STD-105.
4. Any item of required marking information other than 10 through 16 listed under major defects omitted, incorrect, or illegible.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A.* Telephone Repeater TP-14 ( ) shall be preserved and packaged in accordance with the procedures specified for the designated methods as prescribed in Specification MIL-P-116 and described in the following paragraphs:

5.1.1.1 *Technical manual.* Each technical manual shall be packaged Method IC-8 as follows: Inclose each technical manual within a close-fitting, Type II, Class b bag, conforming to Specification MIL-B-117. Heat seal bag to effect closure.

5.1.1.2 *Receiver headset, spare fuses, tubes and vibrator. No packaging required.* Place the spare parts items and the headset within the designated positions of the case and secure with fastenings provided. Close the case and secure all fastenings.

5.1.1.3 *Telephone Repeater TP-14( ) within case.* Each Telephone Repeater TP-14( ) shall be packaged Method IC-5 as follows: Cushion case on all surfaces with cells or pads or both, fabricated of fiberboard having a minimum dry bursting strength of 175 pounds and designed to absorb the shock of impact encountered in handling and transit. Cells and pads shall have a water resistance equal to that of the box in which they are used. Place each cushioned case within a close-fitting, Style RSC, Type CF, Grade 5, fiberboard box, conforming to Specification PPP-B-636. Seal the box as specified in the Appendix of the box specification and in addition, seal all joints and seams with waterproof, pressure sensitive tape, Type III, Class 8, any color, conforming to Specification PPP-T-60. Place the packaged technical manual on the lid of the box and secure in position with waterproof, pressure sensitive tape, Type III, Class 8, any color, conforming to Specification PPP-T-60.

5.1.2 *Level C.* Telephone Repeater TP-14 ( ) shall be packaged in accordance with commercial practice and in a manner that will afford protection against corrosion, deterioration and physical damage during direct shipment to the first receiving activity.

5.2 Packing.

5.2.1 *Level A.* Telephone Repeater TP-14 ( ) shall be packed within a close-fitting, Class 2, Style 4, nailed wood box, conforming to Specification PPP-B-621. Box closure shall be in accordance with the box specification.

5.2.1.1 *Metal strapping.* Shipping containers shall be strapped with flat steel strapping, in conformance with the requirements of the Appendix of the container specification, only for direct shipment to ports.

5.2.2 *Level B.* A quantity of Telephone Repeater TP-14( ) shall be packed within a Class 1, Style 4, nailed, wood box, conforming to Specification PPP-B-621. Fabricate the box to fit the contents snugly. The gross weight of any shipping container shall not exceed 200 pounds. Box closure shall be in accordance with the box specification. No metal strapping is required.

5.2.3 *Level C.* Telephone Repeater TP-14 ( ) shall be packed for shipment in a manner conforming to the requirements of Uniform Freight Classification for rail shipment, National Motor Freight Classification for truck shipment, Parcel Post Regulations, and the regulations of other carriers as applicable to the mode of transportation employed at the lowest transportation rate.

5.3 Marking. Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Standard MIL-STD-129.

6. NOTES

6.1 Intended use. Telephone Repeater TP-

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14( ) is intended for use in extending the telephone communication range of two-wire military line facilities at both terminal and intermediate points. It is capable of operation as a "21" or "22" type repeater.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Type required.

(c) Level of packaging and level of packing required for shipment. (Level A, level B, or level C)

(d) Samples required. (See 3.10)

(e) Marking and shipping of samples.

(f) Place of final inspection.

(g) Technical literature required. (See 3.11)

(h) Quantity of tools and running spare parts required. (See 3.11)

(i) Facsimiles required. (See 3.8)

(j) Preproduction pack(s) as follows:

(1) Makeup of pack(s).

(2) Number of each kind of pack to be submitted.

(3) Inspection that will be performed thereon — including rough handling test, which will not be performed as acceptance inspection.

(4) Five copies of process sheet, using format shown in figure 11 and accompanied by such drawings, sketches, and other data as may be necessary to completely describe the procedure followed in fabrication of the preproduction pack(s) and to identify the items therein. These data should be submitted to the contracting officer with the preproduction pack.

(5) A packaging technician from the procuring agency will evaluate the preproduction pack(s).

6.3 Nomenclature. The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example TP-14-M. As soon as possible, after award of the contract, the contractor should apply to the Government office specified in the contract for such information.

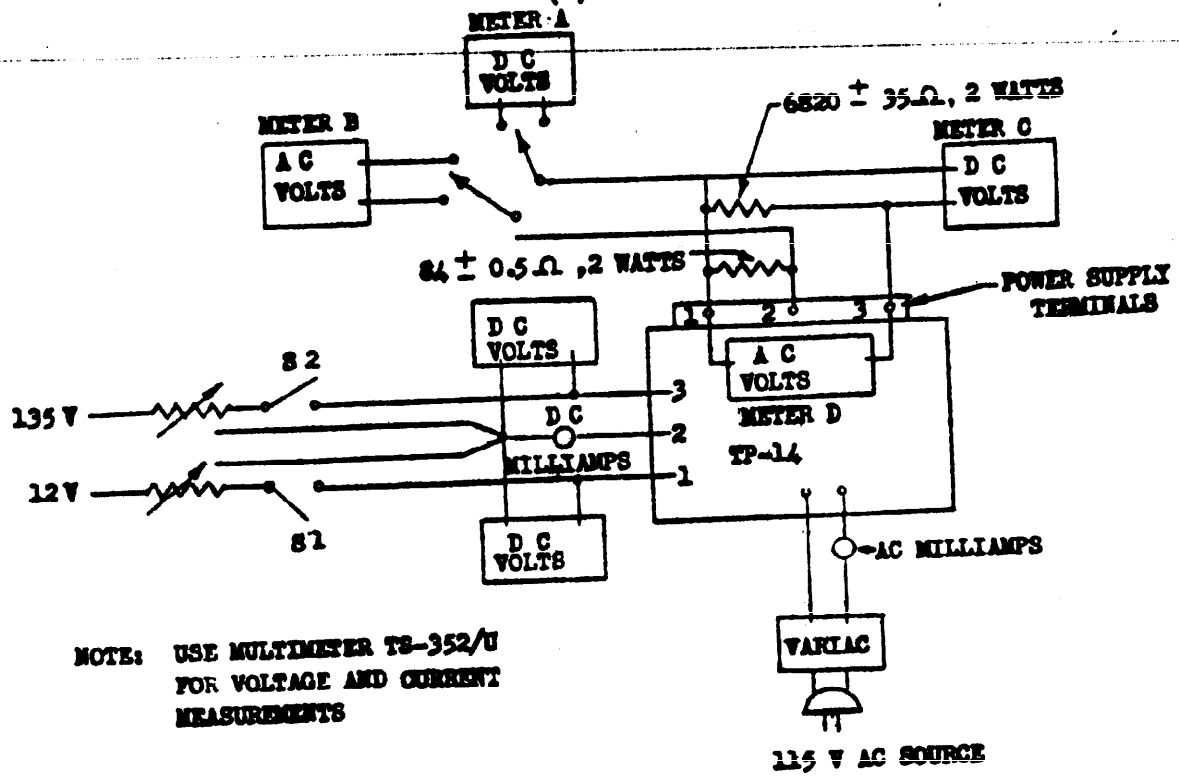
Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever, and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Signal Corps
Air Force

Preparing activity:
Army—Signal Corps

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FIG. 1



NOTE: USE MULTIMETER TS-352/U FOR VOLTAGE AND CURRENT MEASUREMENTS

FIG. 2

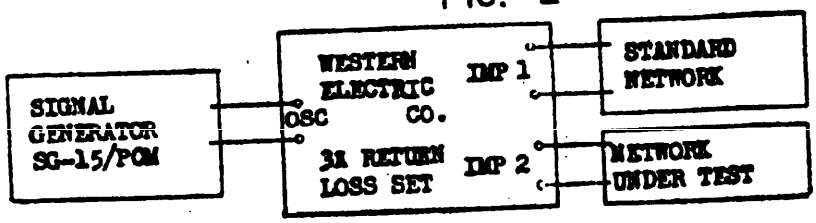
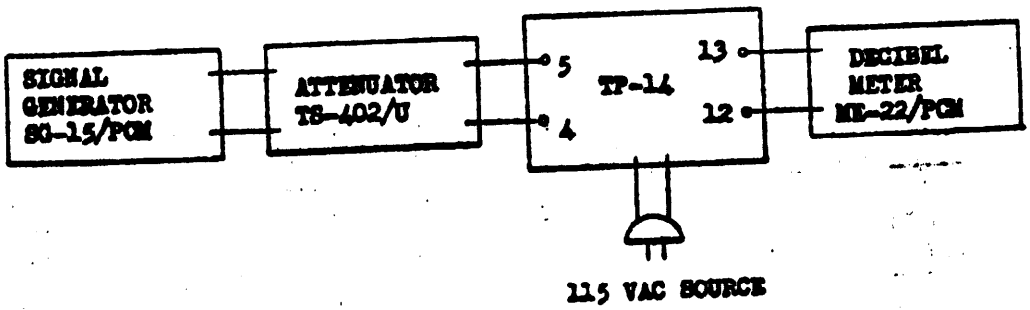


FIG. 3



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TELEPHONE REPEATER TP-14-( )

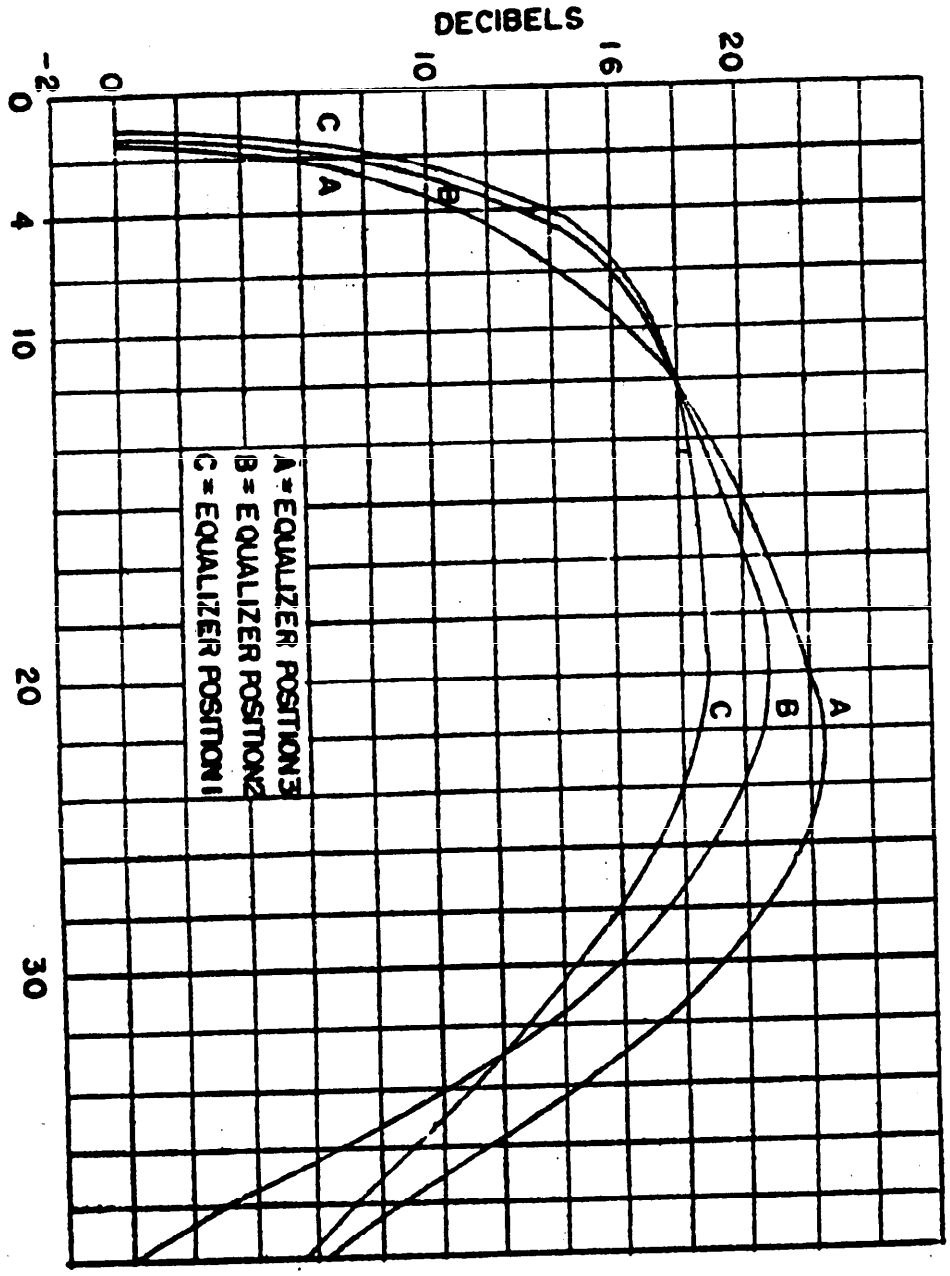


FIG. 4

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FIG. 8

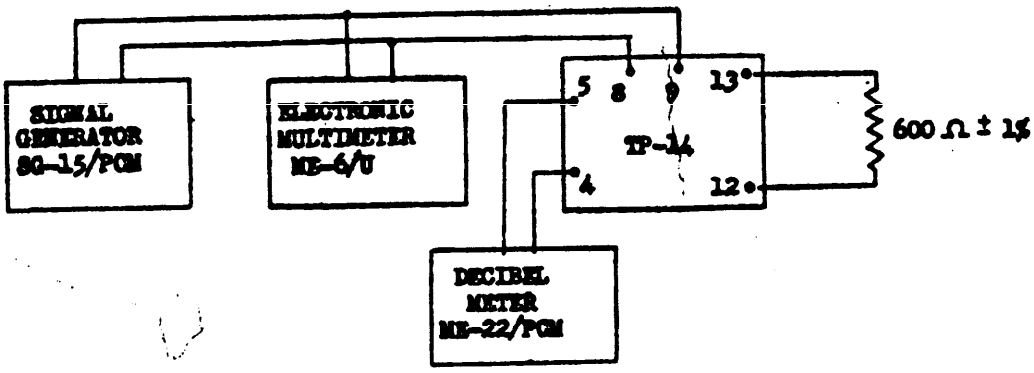


FIG. 9

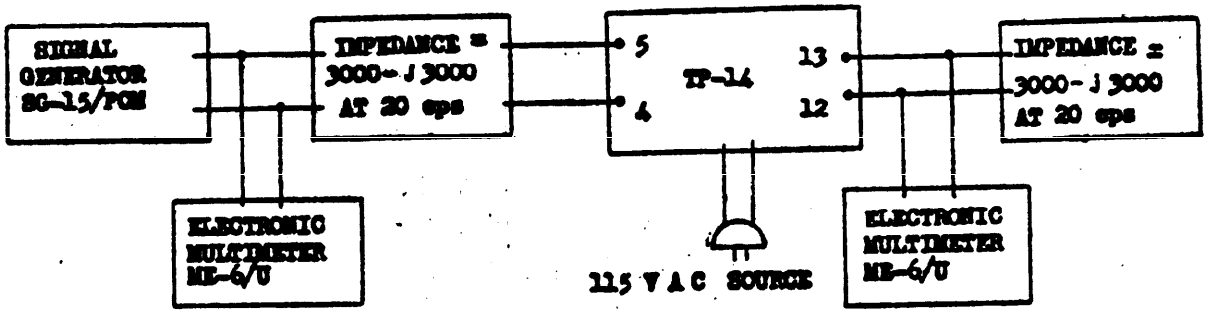
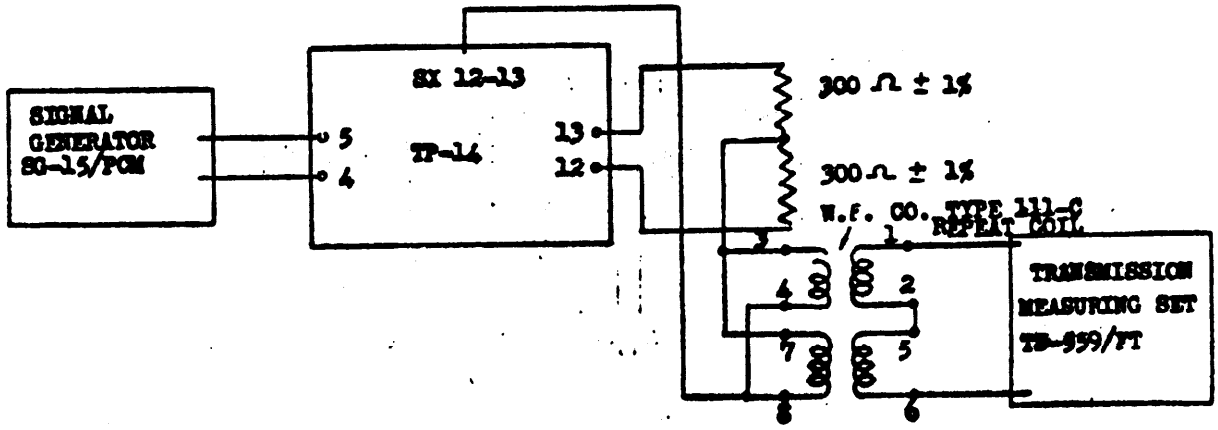


FIG. 10



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FIG. 5

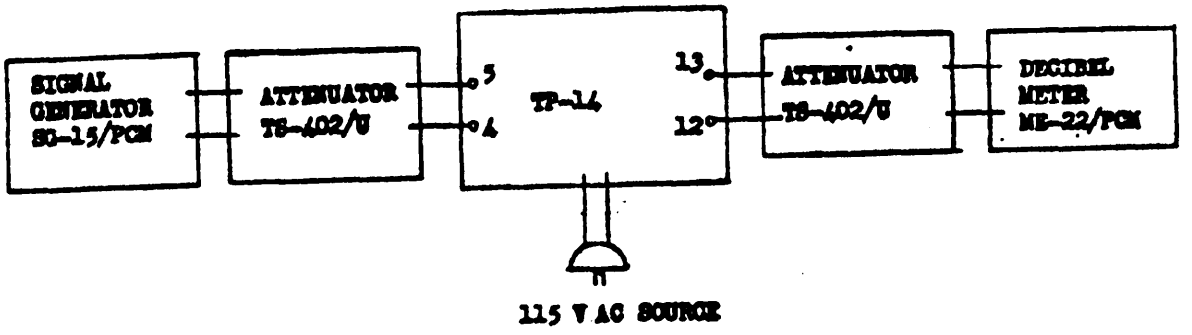


FIG. 6

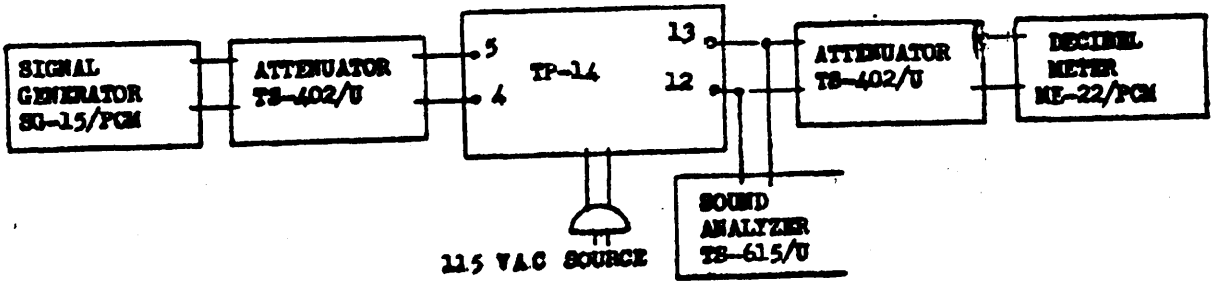
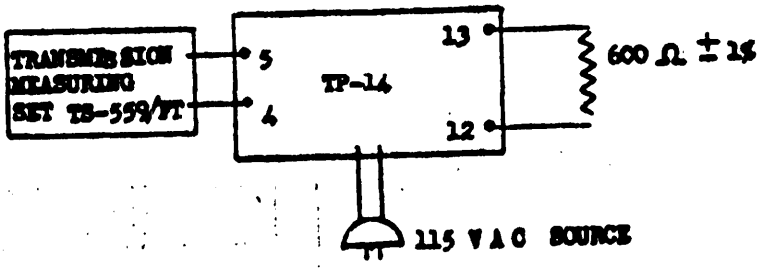


FIG. 7



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PACKAGING, PACKING, AND MARKING PILOT PACK PROCESS SHEET

CONTRACTOR
STREET ADDRESS
CITY AND STATE

ORDER NO.....
EQUIPMENT.....
LEVEL.....

DATE PILOT PACK COMPLETED FOR EVALUATION.....

ITEM NAME			
PACKAGING METHOD			
CLEANING	SPECIFICATION		
	MATERIAL DESCRIPTION	T G C	T G
PRESERVATION	SPECIFICATION		
	MATERIAL DESCRIPTION	T G C	T G
CUSHIONING	SPECIFICATION		
	MATERIAL DESCRIPTION	T G C	T G
	SIZE		
SECURING	SPECIFICATION		
	MATERIAL DESCRIPTION	T G C	T G
	SIZE		
BARRIER	SPECIFICATION		
	MATERIAL DESCRIPTION	T G C	T G
	SIZE		
DESICCANT	SPECIFICATION		
	MATERIAL DESCRIPTION	T G C	T G
	SIZE		
UNIT CARTON	SPECIFICATION		
	MATERIAL DESCRIPTION	T G C	T G
	SIZE		
INTERMEDIATE CARTON	SPECIFICATION		
	MATERIAL DESCRIPTION	T G C	T G
	SIZE		
OUTER CARTON (CONSOLIDATED OR IIb)	SPECIFICATION		
	MATERIAL DESCRIPTION	T G C	T G
	SIZE		
SECURE CLOSURE	SPECIFICATION		
	MATERIAL DESCRIPTION	T G C	T G
	SIZE		

USE AS MANY COLUMNS AS NECESSARY

CASE LINER	SPECIFICATION	
	SEALING MATERIAL	
	SIZE	
SHIPPING CONTAINER	SPECIFICATION	
	STYLE	
	SIZE	
STRAPPING	SPECIFICATION	
	MATERIAL DESCRIPTION	
	SIZE	

LEGEND FOR MATERIAL: T—TYPE, G—GRADE, C—CLASS

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